Ex.no:14 IMPLEMENTING PACKET SNIFFING

08/08/26 USING RAW SOCKETS

AIM:

To implement packet sniffing using RAW sockets in python.

ALGORITHM:

→ Create a raw socket.
→ Bind the socket to the network interface.
→ Receive packets continuously
→ Read the packet header.
→ Display the source & destination IP addr
→ Display the protocol used.
→ continue until the user stops the prgm.

INPUT:

Network packets received through the network interface.

CODE:

```
import socket
s = socket.socket(socket.AF_INET, socket.
                  sock_RAW, socket.IPPROTO_IP)
s.bind(("0.0.0.0", 0))
s.setsockopt(socket.IPPROTO_IP, socket.
                  IP_HDR INCL)
print("Packet Sniffer started...")
while TRUE:
    packet, addr = s.recvfrom(65565)
    ip_header = packet[0:20]
    iph = struct.unpack("!BBHHHBH4s
                        4s", ipheader)
```

```
src_ip = socket.inet_ntoa(iph[8])
dest_ip = socket.inet_ntoa(iph[9])
print("Source IP:", src_ip)
print("Destination IP:", dest_ip)
print("Packet length:", len(packet))
```

OUTPUT:

```
Packet sniffer started...
Source IP: 192.168.1.10
Destination IP: 192.168.1.1
Packet length: 79
```

RESULT:

Thus, a raw socket pgm was successfully implemented to capture and display network packets.